airtel

SYNC

Stadium Surge

5G and 4G plan for a 1,00,000-fan final

₹3.52 Cr, sized from the Airtel case dataset

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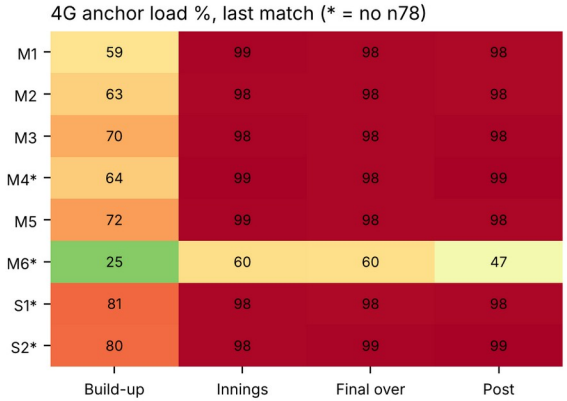
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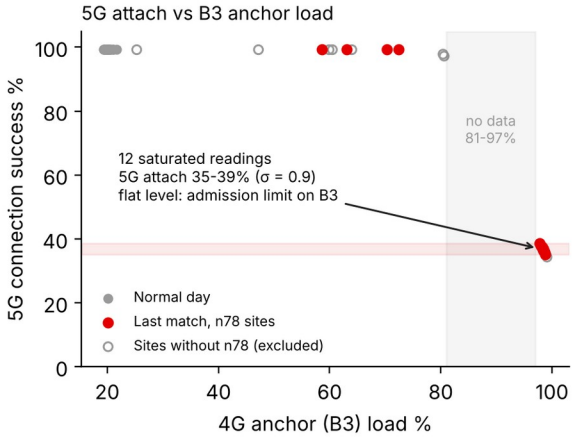


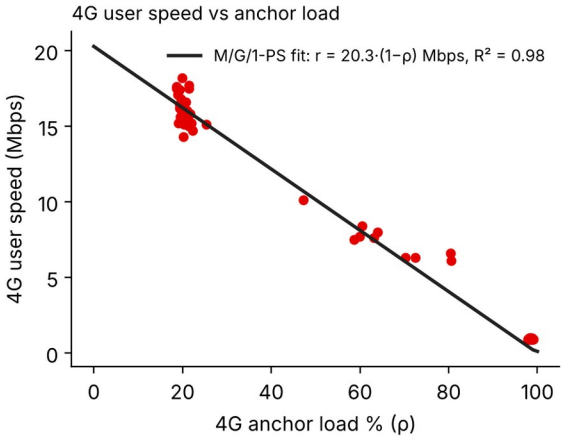
Platform Partner unstop

01 What broke at the last match

Sync_V1.xlsx from the Airtel case brief: 8 sites, 4 match moments, normal day vs the 65k-fan match







35-39%	5G attach in all 12 saturated readings at the four n78 sites (σ 0.9)
99.2%	same four sites at 58-72% anchor load
R² 0.98	4G speed vs load fits r = 20.3(1-ρ) Mbps
24%	expected 5G attach at the final if nothing changes

A flat 35-39% across sites and phases looks like an admission limit on B3 (RRC connected users), since plain congestion would scatter. M4, M6, S1 and S2 have no n78, so their 5G figures are left out. There are no readings between 81% and 97% load, so we plan at 80%.

Phones that could not add 5G stayed on LTE: about 2,100 extra users on a full layer, hence 1 Mbps. Left as is at the final, LTE demand would reach 16.2 Gbps, about ₹5.8 Cr of COWs.

Source: Sync_V1.xlsx in the Airtel Sync 2026 Network case brief (Unstop download)

02 Tasks 1 and 2: demand, cell count, budget

Cells sized at 80% load. 5G massive MIMO is loaded first since it gives the most Mbps per MHz

1,00,000	40,000	28k | 12k	5,040 | 2,160	25.2 | 6.5 Gbps
fans	Airtel (40%)	5G | 4G-only phones	online at peak (18%)	at 5 | 3 Mbps each

Cell type (DL)	Spectrum × SE	@80%	Users
5G macro 64T64R mMIMO	100 MHz × 74% × 16	947 Mbps	189 @5
5G small cell 4T4R	100 MHz × 74% × 7	414 Mbps	83 @5
4G macro sector	70 MHz × 2.5	140 Mbps	46 @3
4G small cell	40 MHz × 3.0	96 Mbps	32 @3

Requirement @80%	5G	4G
Peak demand	25.2 Gbps	6.5 Gbps
Cells needed (macro-eq.) / existing	27 / 12	47 / 20
Extra cells needed	**+15**	**+27**

Item	Qty	Cost	Adds
Under-seat small cells (4G + 5G)	24 × ₹8 L	₹1.92 Cr	9.9 Gbps 5G, 2.3 Gbps 4G
Cells on wheels (gates, M4/M6 side)	4 × ₹25 L	₹1.00 Cr	12 sectors each on 4G and 5G
Fibre ring, 2 × 10G	1.2 km	₹0.60 Cr	backhaul for new cells
Reserve: licences, dry run, spare cells	-	₹0.48 Cr	up to 6 more small cells
Total		**₹4.00 Cr**	

Capacity check @80% (Gbps)	5G	4G
Existing + 4 COWs + 24 small cells	32.7	6.8
Peak load at the final	**62%**	**76%**

Peak load of our mix (5G / 4G)	SE as planned	SE 25% lower	SE 40% lower
15% of Airtel users online	51% / 64%	69% / 85%	86% / 106%
18% online (case value)	62% / 76%	82% / 102%	103% / 127%
25% online	86% / 106%	114% / 142%	143% / 177%

4G is the tight layer

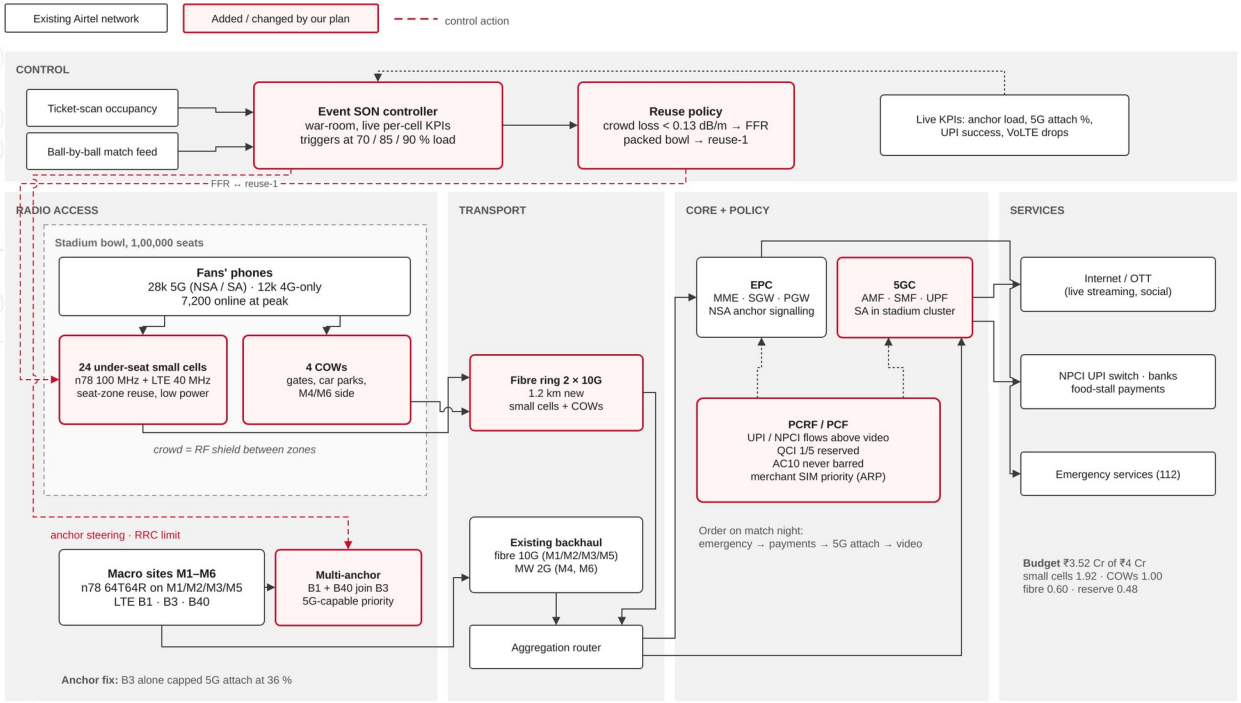
It fills at about 24% online. The D-30 dry run measures real SE and load; if either is off, the reserve buys up to 6 more small cells and the 720p video cap starts at 85% load.

Source: fan numbers, targets, spectrum and unit costs from the case brief. Spectral efficiency values are our planning assumptions

03 Network changes for match night

Black outline: existing. Red outline: new or changed. Dashed red: control actions

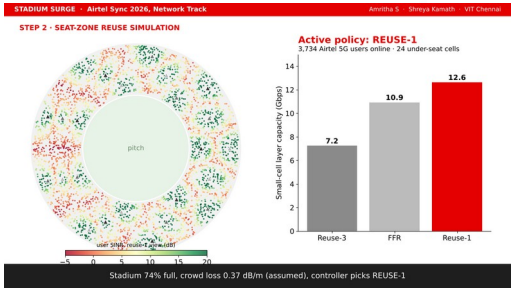
Stadium Surge: network architecture for the final



How it fits together

- SON controller takes ticket-scan counts, the ball-by-ball feed and live cell KPIs
- Under-seat cells and COWs connect to a new 2 × 10G fibre ring
- B1 and B40 become NSA anchors next to B3
- SA core for the stadium cluster, PCF rules for UPI and 112

[Demo video](#) | [GitHub repo](#)



04 Task 3: 5G attach, payments, emergency calls

B3 ran out of RRC room, SgNB addition failed, and 5G phones stayed on LTE

Getting 5G phones onto 5G

- B1 and B40 as extra anchors
- Keep anchor capacity for EN-DC phones, move LTE-only phones to other bands (SPID/RFSP)
- Lower B1 threshold for SgNB add, no SgNB release on anchor load
- Higher RRC licence and a shorter inactivity timer on match day
- SA on n78 for phones that support it
- Steer load to M6 (60% at peak) with tilt and mobility offsets
- Target: 95%+ attach (was 36%)

UPI payments

- Small packets, but time critical. PCC rule for NPCI and bank endpoints, ahead of video
- Stall merchant SIMs on a higher ARP profile, small cells near the food courts
- Backup: UPI Lite, UPI 123PAY, stall Wi-Fi
- Target: under 1.5% failures (was 7.1%)

112 and VoLTE

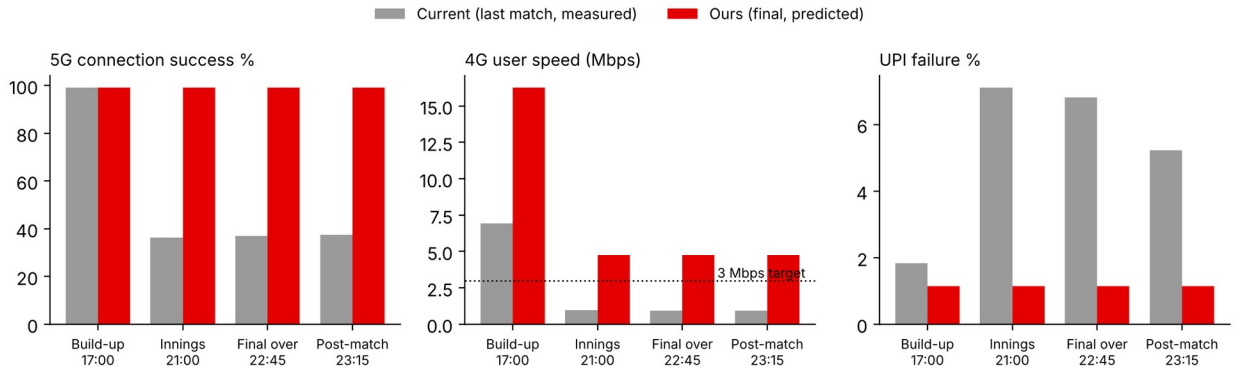
- AC10 and QCI 1 never barred, pre-emption allowed
- About 5% PRB and RRC headroom kept for QCI 1/5
- AC 11-15 SIMs for police, medical and ground staff
- Barring for normal users only past 90% load
- VoNR on SA, so voice does not fall back to LTE
- Target: every emergency call sets up

Priority if a cell fills up: 112, then UPI, then 5G attach, then video.

Source: EN-DC (NSA) procedures: 3GPP TS 37.340

05 Current vs proposed

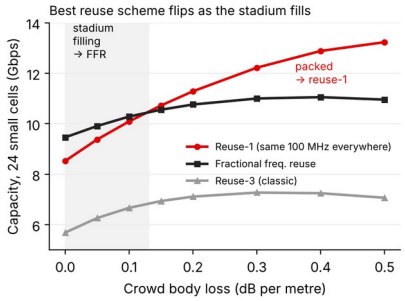
Grey: last match, from the case data. Red: predicted for the final with the same fitted models. Below: our simulations

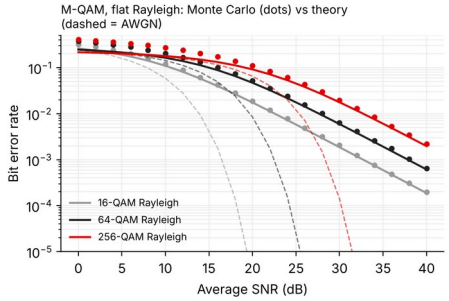


Seat-zone reuse

At 3.5 GHz a dense crowd adds body loss, which TR 38.901 models as blockage. Under-seat cells end up isolated from each other, so all 24 can run the full 100 MHz.

With the stands half empty FFR does better, so the controller changes scheme with occupancy.





Simulation results

- Full stadium, 24 cells: 12.2 Gbps with reuse-1, 11.0 with FFR, 7.3 with reuse-3
- Crossover near 0.13 dB/m crowd loss
- Rayleigh M-QAM Monte Carlo matches theory. 256-QAM needs about 25-30 dB
- 5 seeds, 5,040 users. Same code in MATLAB and Python

Source: body blockage: 3GPP TR 38.901 sec. 7.6.8 | SINR to throughput: 3GPP TR 36.942 Annex A.2 | crowd loss per metre is our assumption, not yet measured

06 Task 4: user experience side effects

Mostly from adding 28 cells in and around one bowl

Possible impact	Cause	Mitigation
Interference, lower SINR	Many cells in a small area	Low-power under-seat antennas, downtilt, PCI/PRB plan, ICIC between small cells
Handover ping-pong, dropped calls	Fans moving between macro, COW and small cell	Hysteresis and time-to-trigger tuning, MRO, checked at the dry run
More users online than planned	A faster network pulls more demand	Reserve becomes extra small cells; 720p cap only above 85% load
Uneven speed, stands vs concourse	Small-cell placement gaps	Seat-level walk tests at the dry run, fix placement before D-14
Short access delay	Barring for normal users	Only above 90% load, short barring timers
Changes leak outside the venue	Re-tilted macros, COW coverage	Event profile switched on 3 h before; COW power capped; all restored after
Post-match crush at gates and metro	Traffic moves from bowl to exits	2 COWs at the exits
Battery drain, voice fallback delay	NSA dual connectivity	SA drops the LTE leg, DRX tuning, VoNR

Match-day monitoring

Per-cell dashboard (anchor load, 5G attach, UPI success, VoLTE drops) with actions at 70, 85 and 90% load. Dry run at a league match 30 days out, with rollback for every parameter. Extra capacity goes in 10 minutes before the innings break and the last over.

07 Competitive analysis and impact

Typical ways to add venue capacity, judged against this case (60 days, ₹4 Cr)

	COW-heavy	Permanent in-venue DAS	5G SA only	Our plan
Fixes 5G attach failure	No	No	Yes, for SA phones	Yes: multi-anchor + SA
Capacity per seat	Medium, interference limited	High	Depends on radio layer	High: reuse-1 when packed
Ready in 60 days	Yes	No, months of civil work	Needs core changes	Yes
Cost profile	₹25 L every event	High capex	Core and device dependent	₹3.52 Cr, 73% reusable
Protects UPI and 112	Not addressed	Not addressed	Possible with slicing	PCC rules + barring policy
Sized from data	Rule of thumb	Venue survey	-	Fitted on case data (R^2 0.98)

36% to 99%	1 to 4.8 Mbps	12 to 70+ Mbps	7.1% to 1.2%	all set up
5G attach	4G speed	5G speed	UPI failure	112 calls

Timeline: D-60 survey, orders | D-45 fibre, small cells | D-30 dry run at a league match | D-14 change freeze | D-7 COWs in place | D+1 review

Within ₹4 Cr, about 40,000 Airtel fans get 5G that attaches, 4G above target, and payments that go through.

Code, MATLAB/Simulink and demo: github.com/Amritha902/airtel-sync-stadium-surge | demo video